

# Read me first — sim re-entry (540, July 2026)

Last synced: 2026-07-27 —  $\rho$  assortativity, archive manifest, Pass B ablation spec.

OPORD: `../3-Master_Plan/re_entry/model_OPORD.md`

If you feel lost: read `../3-Master_Plan/re_entry/00_READ_ME_FIRST.md` (docs 01–03) before opening sim code.

## Revolution vs evolution

| Do                                                                                                                           | Don't                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Import <code>tier1_pool_assignment.py</code> ,<br><code>tier1_generative_eda.py</code> ,<br><code>tier1_sim_config.py</code> | Extend archived notebooks ( <code>538D</code> , <code>538</code> , <code>537</code> , <code>535</code> ) |
| Thin scripts / <code>540_*</code> notebook                                                                                   | Copy-paste 538D cells                                                                                    |
| Answer Alex-sized questions (knockouts, ablations)                                                                           | Rivanna sweeps, bin-for-bin hero match as v1 gate                                                        |

Archived notebooks in `sports/archive/` are **SCOUT-era lab reference** — read only, do not edit.

## Charles’s three-step pipeline

- 1. **Assign** — build rosters ( $\rho$  or sort-and-chop)
- 2. **Score** —  $S_i = A_i - \lambda \cdot L_C$  (Alex v1:  $(B-D)=-L_C$  in selection)
- 3. **Draft** — top K by  $S_i$

Knockouts (separate experiments):

| Pass | What toggles                                                   | Status                                    |
|------|----------------------------------------------------------------|-------------------------------------------|
| A    | $\lambda=0$ vs congestion in $S_i$                             | Done → <code>alex_side_by_side_v0/</code> |
| B    | $\rho$ low / $\rho$ high / sort-and-chop; <b>fix</b> selection | <code>540_rho_ablation_bundle.py</code>   |

## Three-step assignment (step 1 only)

- 0. Draw latent abilities **A<sub>i</sub>**
- 1. Draw team targets **T<sub>j</sub>** (e.g. Uniform on [−0.5, 0.5])
- 2. Sequential soft match with roster caps:

$$\pi_{ij} \propto \exp \left( -\rho \cdot \frac{(A_i - T_j)^2}{2\sigma^2} \right) \times (n_j + k)^\alpha$$

Optional preferential attachment: `USE_PREFERENTIAL_ATTACHMENT = True` → **α > 0**; default **False**.

| ρ             | Meaning                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------|
| ρ = 0         | Uniform among teams with roster room (max <b>mixing</b> )                                               |
| ρ = 1         | Moderate assortativity (≡ legacy <b>τ = σ ≈ 0.65</b> )                                                  |
| ρ high        | Sharper match to nearest <b>T<sub>j</sub></b>                                                           |
| sort-and-chop | Separate code path — 537-style global sort + equal slices; <b>max assortativity benchmark</b> , not ρ→∞ |

**σ** (`ASSIGNMENT_SIGMA`), default **0.65**) is a fixed scale — not the user-facing knob.

Legacy docs use **τ (temperature)** with **opposite** intuition (small τ = assortative). **540+ uses ρ only** in prose and exports.

## Hero estimand lock

| Field       | Value                                          |
|-------------|------------------------------------------------|
| Outcome     | <code>Y_draft</code> , mean draft rate per bin |
| X (hero)    | <code>poolq_loo</code>                         |
| Bins        | 16 quantile                                    |
| Winsor      | (0.01, 0.99)                                   |
| min_minutes | 20                                             |
| Seasons     | 2011–2021                                      |

Slug: `empirical_ppm_poolq_loo_16quantile_winsor0199_min20_2011`

## Active files

| Role          | Path                                                                     |
|---------------|--------------------------------------------------------------------------|
| Empirical     | 530_sports_pipeline.ipynb , sports_pipeline/                             |
| Sim engines   | tier1_pool_assignment.py , tier1_generative_eda.py , tier1_sim_config.py |
| Pass A bundle | scripts/hero_model_reset_bundle.py → alex_side_by_side_v0/               |
| Pass B bundle | scripts/540_rho_ablation_bundle.py → alex_rho_ablation_v0/               |
| Notebook      | 540_three_step_sim.ipynb                                                 |
| Binding       | 3-Master_Plan/BINDING_Selection_is_its_own_step.md                       |
| Slide         | 3-Master_Plan/re_entry/Model.pdf                                         |

## Export naming

```
sports/datasets/mbb/exports_inverted_u_v0/  
alex_side_by_side_v0/      # Pass A (λ knockout)  
alex_rho_ablation_v0/     # Pass B (ρ ablation)
```

Pass B artifacts: generative\_rho\_low\_16quantile.csv , generative\_rho\_high\_16quantile.csv , generative\_sort\_chop\_16quantile.csv , rho\_ablation\_summary.txt , rho\_ablation\_caption.txt , PNG.

## Anti-patterns

- Opening archive/ notebooks as the daily workspace
- Merging L\_net and S\_i in comments or claims
- Saying ρ ablation proves hero bin-for-bin match
- Using τ in new user-facing text (archive reference only)

# When stuck

| Question                         | Read                                                                                                           |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|
| What are we proving?             | <code>re_entry/02</code> , <code>Model.pdf</code>                                                              |
| Assignment math (legacy $\tau$ ) | <code>documents/Tier1_Presorting_Design_Note.md</code>                                                         |
| Widget-era manual                | <code>documents/538_Cell10_Generative_Manual.md</code><br>(archive context)                                    |
| Implementation                   | <code>tier1_pool_assignment.py</code> — <code>soft_assign</code> ,<br><code>simulate_generative_rosters</code> |